Mission 5: The Stroke Unit

The story

"10:00 at Qendra Neuro Tirana. Arben arrives, but the scanner is busy with another patient, and two more ambulances are on the way. Mission 4 assumed the hospital would be waiting for him. Real mornings are not like that. Your fifth mission simulates one morning at the hospital and decides who goes into the scanner next."

What you need to know

- Qendra Neuro Tirana has **one CT scanner** and **one clot removal room**.
- Every patient is scanned first. The scanner handles one patient at a time.
- After the scan, a patient who can still get the drug gets it. A patient who needs clot removal (big pipe and severity 5 or more, as in Mission 3) then goes to the clot removal room. The room handles one patient at a time.
- **Door-to-drug** is the number of minutes from arrival to the drug. Hospitals aim for 60 minutes or less.

Use these fixed times:

Step	Minutes
A scan takes	15
From the end of the scan to the drug	15
From the end of the scan until the clot removal room can start	45
Clot removal takes	60

With an empty hospital, these give exactly the Mission 4 times: the drug 30 min after arrival, clot removal 60 min after arrival. Every minute more is time spent waiting in line.

The base challenge

Your goal: replay one morning at the hospital and show who waited, for how long, and who missed the drug because of it.

Goal 1: Read the day's patients

The patients of the morning come from two places:

1. `data/today.events`: patients from other parts of the country. They are not in your database.
2. Your database: every case whose saved route (Mission 4) ends at Qendra Neuro Tirana. Its arrival time, severity, last seen well and "clot removal possible" come from the case.

`data/today.events` uses the same format as the other files:

```
# ARRIVAL | time | patient | type | severity | last seen well | big pipe

ARRIVAL | 09:20 | P-0037 | stroke | 5  | 06:10 | no
ARRIVAL | 09:40 | P-0039 | stroke | 6  | 08:10 | no
ARRIVAL | 09:55 | P-0040 | stroke | 3  | 09:05 | no
ARRIVAL | 10:05 | P-0043 | stroke | 12 | 09:30 | yes
ARRIVAL | 10:25 | P-0044 | stroke | 9  | 06:30 | no
```

With the cases from Mission 4, the morning has 8 patients: these 5, plus P-0001 (Arben, 10:00), P-0002 (Drita, 08:30) and P-0004 (Mira, 11:50). P-0003 goes to Shkoder, so he is not here.

Goal 2: Simulate the morning

`unit run <file>` runs the morning with these rules:

1. A patient joins the waiting line at their arrival time.
2. Whenever the scanner is free and someone is waiting, it takes the next patient: **first come, first served**. Same arrival time: the smaller id first.
3. The drug is given 15 min after the scan ends, if that time is within the patient's drug limit (last seen well + 4 h 30 min). Otherwise the patient misses the drug.
4. A patient who needs clot removal is ready for the room 45 min after the scan ends. The room takes the patient who was ready first, as soon as it is free.

Print every event in time order, then one summary row per patient. Mark door-to-drug over 60 min as `SLOW`.

Example CLI interaction (you may design yours differently):

```
synapse> unit run data/today.events
Qendra Neuro Tirana · 1 CT scanner · first come, first served
8 patients: 5 from today.events, 3 from your cases (P-0003 goes to another hospital)

08:30  P-0002 arrives
08:30  P-0002 scan starts
09:20  P-0037 arrives
09:20  P-0037 scan starts
09:30  P-0002 clot removal starts
09:40  P-0039 arrives
09:40  P-0039 scan starts
09:50  P-0037 drug given (30 min after arrival)
09:55  P-0040 arrives
09:55  P-0040 scan starts
10:00  P-0001 arrives
10:05  P-0043 arrives
10:10  P-0001 scan starts
10:10  P-0039 drug given (30 min after arrival)
10:25  P-0044 arrives
10:25  P-0043 scan starts
10:25  P-0040 drug given (30 min after arrival)
10:40  P-0044 scan starts
10:40  P-0001 drug given (40 min after arrival)
10:55  P-0043 drug given (50 min after arrival)
11:00  P-0044 drug limit passed before the drug could be given
11:10  P-0001 clot removal starts
```




```
11:50  P-0004 arrives
11:50  P-0004 scan starts
12:10  P-0043 clot removal starts
13:10  P-0004 clot removal starts

PATIENT  ARRIVES  SCAN   DRUG   DOOR-TO-DRUG  CLOT REMOVAL  NOTE
P-0002   08:30    08:30  -      -             09:30         drug limit passed before arrival
P-0037   09:20    09:20  09:50  30 min        -
P-0039   09:40    09:40  10:10  30 min        -
P-0040   09:55    09:55  10:25  30 min        -
P-0001   10:00    10:10  10:40  40 min        11:10
P-0043   10:05    10:25  10:55  50 min        12:10
P-0044   10:25    10:40  -      -             -             MISSED THE DRUG (limit 11:00)
P-0004   11:50    11:50  -      -             13:10         drug limit passed before arrival

Drug given to 5 of 6 patients who could get it. Average door-to-drug: 36 min.
SUCCESS: Hospital times saved for P-0001, P-0002, P-0004.
```

P-0044 arrived with 35 minutes left on the drug limit, waited behind P-0043, and lost it. Bonus 1 is about that problem.

Goal 3: Save the results

For every patient from your database, save the hospital times in the case. Running the morning again replaces them.

Example of the saved times inside the case (you may design yours differently):

```
"hospitalStay": {
  "hospital": "Qendra Neuro Tirana",
  "arrives": "10:00",
  "scanAt": "10:10",
  "drugAt": "10:40",
  "doorToDrugMinutes": 40,
  "clotRemovalAt": "11:10"
}
```

Error handling

Your program must handle and clearly report errors such as:

- "ERROR: File not found: data/today2.events"
- "ERROR: today.events, line 3: invalid time "9:2O". Use HH:MM."
- "ERROR: today.events, line 5: unknown patient type "strok". Expected stroke."
- "ERROR: today.events, line 6: severity must be a whole number, found "high"."
- "ERROR: today.events, line 7: big pipe must be yes or no."
- "ERROR: today.events, line 8: patient P-0039 is listed twice (first on line 4)."
- "Case P-0005 has no route yet. Run route P-0005 first. Skipped." (a message, the morning still runs)
- "No patients arrive this morning." (a message, not a crash)

The bonus challenge

For extra points.



Bonus 1: Deadline first

First come, first served let P-0044 miss the drug. Add `--order deadline`: the scanner takes the patient whose **deadline** comes first.

- For a patient who can still get the drug: deadline = drug limit − 30 min (the last moment the scan can start).
- For everyone else: deadline = arrival + 60 min.
- Same deadline: higher severity first, then earlier arrival, then the smaller id.

```
synapse> unit run data/today.events --order deadline
...
PATIENT  ARRIVES  SCAN   DRUG   DOOR-TO-DRUG  CLOT REMOVAL  NOTE
...
P-0001   10:00    10:10  10:40  40 min        11:10
P-0043   10:05    10:40  11:10  65 min SLOW   12:10
P-0044   10:25    10:25  10:55  30 min        -
...
Drug given to 6 of 6 patients who could get it. Average door-to-drug: 38 min.
```

P-0044 is saved, but P-0043 now waits 65 minutes. There is no free lunch with one scanner.

Bonus 2: One scanner or two?

`unit compare <file>` runs the same morning with 1 and with 2 scanners and prints both results side by side.

```
synapse> unit compare data/today.events
                        1 SCANNER   2 SCANNERS
Drug given              5 of 6      6 of 6
Average door-to-drug    36 min      31 min
SLOW (over 60 min)      none        none
Missed the drug         P-0044      none
```

Bonus 3: Bus crash

Later in the day we hand out `data/today-v2.events`. It adds four head injuries from a bus crash near Tirana:

```
ARRIVAL | 09:58 | H-01 | head | - | - | -
ARRIVAL | 09:58 | H-02 | head | - | - | -
ARRIVAL | 09:58 | H-03 | head | - | - | -
ARRIVAL | 09:58 | H-04 | head | - | - | -
```

A head injury needs a scan within 60 min of arrival. It never gets the drug or clot removal. Report every head injury scanned late. The results to expect:

Order	Scanners	Missed the drug	Head scans late	Arben's door-to-drug
First come	1	P-0044	none	100 min
Deadline first	1	none	H-04	115 min
Deadline first	2	none	none	70 min



Hints

- Play the example morning on paper first, patient by patient. Your program must make the same choices you made.
- Several things can happen in the same minute: an arrival, the end of a scan, a drug. Decide in which order your program handles them, and check it against the example.
- Bonus 1 changes only who goes into the scanner next. Think about how much of your code that should touch.

What the judges will test

- The morning above, then hidden mornings.
- Two patients arriving in the same minute, and a patient arriving the exact minute the scanner becomes free.
- An empty file, and a case with no route.
- `today-v2.events` with the bus crash.

What this mission saves

The hospital times inside every case that went to Qendra Neuro Tirana. Mission 7 prints them next to the Mission 4 estimates, so a doctor can see how long the patient waited.